LESSON 6.1

Division Expressions with Fractions and Whole Numbers

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can explain what dividing with a whole number and a unit fraction means — how many groups fit, or how much is in one group — find the quotient with a model, and check it by multiplying.

WHY THIS MATTERS Agent Reyes finds a 3-foot strip of tape at the crime scene. The lab needs the tape cut into 1/4-foot sections for analysis. How many sections can be made? The answer cracks the code to the locked evidence vault.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Reciprocal *Spanish: Recíproco*	A fraction turned upside down.	1/4 → 4/1
Quotient *Spanish: Cociente*	The answer when you divide.	8 ÷ 1/2 = 16 is the quotient.
Dividend *Spanish: Dividendo*	The total number being divided into equal groups (the number inside the division bracket).	In 8 ÷ 1/2, 8 is the dividend.
Divisor *Spanish: Divisor*	The number of equal groups you are dividing into (the number outside the division bracket).	In 8 ÷ 1/2, 1/2 is the divisor.
Unit fraction *Spanish: Fracción unitaria*	A fraction with 1 on top, like 1/4.	1/3, 1/8, 1/10
Multiplicative inverse *Spanish: Inverso multiplicativo*	Two numbers whose product is 1; each is the reciprocal of the other.	4 and 1/4 · 2/3 and 3/2

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A cook has 4 cups of broth. How many 1/3-cup servings can she pour?

1. Write the division: 4 ÷ 1/3.
2. Keep 4, change ÷ to ×, flip 1/3 to 3: 4 × 3 = 12.
3. Check: 12 × 1/3 = 4 cups.

Answer: 12 servings

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Let's try 2 ÷ 1/3. The bar model counted 6 thirds inside 2 wholes — now let's see why.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- To divide by a fraction, I multiply by its _____ .
- I wrote _____ over 1 so both numbers are fractions.
- The quotient is _____ because that many groups fit inside.
- Dividing by a fraction smaller than one whole makes the quotient _____ .

Word bank reciprocal · quotient · dividend · divisor · unit fraction · keep · change · flip · multiply · groups · fit inside · larger

WATCH OUT!

A common mistake

A common mistake in Interpret Division of Fractions is assuming that dividing always makes the answer smaller. For 4 ÷ 1/3, students reason 'dividing shrinks the number' and compute 4 ÷ 3 ≈ 1.3 instead of asking how many 1/3-foot pieces fit into 4 feet (each whole foot holds 3 thirds, so 4 × 3 = 12). Before you submit, check: if the divisor is a fraction smaller than 1, the quotient must be BIGGER than the dividend, not smaller.

LESSON 6.1

Division Expressions with Fractions and Whole Numbers

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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WHY THIS MATTERS Agent Reyes finds a 3-foot strip of tape at the crime scene. The lab needs the tape cut into 1/4-foot sections for analysis. How many sections can be made? The answer cracks the code to the locked evidence vault.

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

What does 3 ÷ 1/4 mean?

(A) How many 1/4-size pieces fit into 3
(B) 1/4 of 3
(C) 3 groups of 1/4
(D) 3 minus 1/4

HINT Re-read the expression: 3 is the total amount and 1/4 is the size of one piece. What question could that setup be asking?

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Which expression means 'how many 1/3-size pieces are in 2'?

(A) 1/3 ÷ 2
(B) 2 ÷ 1/3
(C) 2 × 1/3
(D) 2 + 1/3

HINT Find the two jobs: which number is the total being split, and which is the size of each piece?

SHOW YOUR WORK

PART 2 Practice the skill

show every step

3 **Circle the letter of the best answer. Show how you know.**

What is 2/3 ÷ 4?

(A) 1/6

(B) 2/7

(C) 3/8

(D) 8/3

HINT The whole number is second, so write the 4 as 4/1.

SHOW YOUR WORK

PART 3 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

A rope is 4 feet long. How many 1/2-foot pieces can be cut from it?

(A) 1/2 piece

(B) 2 pieces

(C) 4 pieces

(D) 8 pieces

HINT Note the numbers: the rope is 4 feet long, and every piece is 1/2 foot — smaller than a whole foot.

Write any whole number over 1	_____
Keep · Change · Flip	_____
Multiply across	_____
Simplify · label the units	_____

1. **Whole over 1** *3 becomes 3/1.*
2. **Keep** *The first fraction does not move.*
3. **Change · Flip** *÷ becomes ×; flip the second fraction.*
4. **Check** *Dividing by a piece smaller than 1 gives a BIGGER answer.*

5 **Circle the letter of the best answer. Show how you know.**

A half-pan of brownies is shared equally among 4 people. How much of the whole pan does each get?

(A) 1/8
(B) 1/2
(C) 2
(D) 4/2

HINT Draw half a pan, then split that half into 4 equal parts.

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Agent Cole has 3/4 of a pound of fingerprint powder and splits it equally into 2 kits. How much powder goes in each kit?

(A) 1 1/2 pounds
(B) 1/8 pound
(C) 3/8 pound
(D) 3/4 pound

HINT Draw 3/4 of a bar, then cut each of those three fourths in half.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Why does dividing by a unit fraction always give a number larger than the dividend?

SENTENCE STARTER Dividing by 1/3 means finding how many ___ fit into ___, so the quotient is ___ than the dividend.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 6.1

Division Expressions with Fractions and Whole Numbers

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each division expression to its quotient.

_____	5 ÷ 1/2	(A)	10
_____	6 ÷ 1/3	(B)	18
_____	2 ÷ 1/6	(C)	12
_____	4 ÷ 1/4	(D)	16
_____	3 ÷ 1/5	(E)	15
_____	7 ÷ 1/2	(F)	14

2 **Use the model to solve. Show your work.**



How many 1/6-size pieces fit into 3 wholes?

SHOW YOUR WORK

Answer: _____

3 **Mark and label each value on the number line.**

Place each result on the number line



SHOW YOUR THINKING

4 **Circle the letter of the best answer. Show how you know.**

A shelf is 5 feet long. Books are each 1/4 foot wide. How many books fit on the shelf? Which expression represents this?

(A) 1/4 ÷ 5 = 1/20 book

(B) 5 ÷ 1/4 = 20 books

(C) 5 × 1/4 = 1 1/4 books

(D) 5 + 1/4 = 5 1/4 books

SHOW YOUR WORK

5 **Write the letter of the matching item on each line.**

Match each fraction ÷ whole number expression to its quotient.

_____ 3/4 ÷ 2

_____ 2/3 ÷ 4

_____ 5/6 ÷ 2

_____ 3/5 ÷ 3

_____ 4/9 ÷ 2

_____ 5/8 ÷ 5

(A) 3/8

(B) 1/6

(C) 5/12

(D) 1/5

(E) 2/9

(F) 1/8

PART 2 Apply it

real-world problems

6 **Circle the letter of the best answer. Show how you know.**

A 5/6-mile stretch of the evidence trail is split equally among 3 search teams. How much of a mile does each team walk?

(A) 1/18 mile

(B) 5/2 miles

(C) 5/6 mile

(D) 5/18 mile

Write any whole number over 1 _____

Keep · Change · Flip _____

Multiply across _____

Simplify · label the units _____

EXPLAIN YOUR THINKING

Why does dividing by a unit fraction always give a number larger than the dividend?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 6.1

Division Expressions with Fractions and Whole Numbers

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Complete the pattern: as the unit fraction divisor gets smaller, what happens to the quotient?

Expression	Quotient	Pattern
2 ÷ 1/2		
2 ÷ 1/3		
2 ÷ 1/4		
2 ÷ 1/6		
2 ÷ 1/10		

SHOW YOUR WORK

2 **Complete the table. Show how you found each value.**

Complete the pattern: the same 3/4 shared among more and more people.

Expression	Quotient	Pattern
3/4 ÷ 2		
3/4 ÷ 3		
3/4 ÷ 4		
3/4 ÷ 6		

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

3 **Find the mistake. Explain it, then show the correct work.**

Find the Interpretation Error

1. **Problem** Interpret 4 ÷ 1/5
2. **Student says** 4 ÷ 1/5 means 1/5 of 4
3. **Student's answer** 4/5

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

4 **Answer in complete sentences.**

Dividing 6 by 1/2 gives 12, but dividing 6 by 2 gives 3. How can dividing by a smaller number (1/2) give a bigger answer than dividing by a larger number (2)? Explain using a real-world example.

WRITE YOUR ANSWER

5 **Find the mistake. Explain it, then show the correct work.**

Spot the Mistake — When the Numerator Is Not 1

1. **Problem** A 3/4-pound bag of powder is split equally into 2 kits. Find 3/4 ÷ 2.
2. **Student writes** The numerator is not a 1, so I will divide straight across: 3 ÷ 2 on top and 4 ÷ 2 on the bottom.
3. **Student's answer** 1.5/2 of a pound

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Why does dividing by a unit fraction always give a number larger than the dividend?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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